TWO FACTOR AUTHENTICATION:

What is two factor authentication????????

- Two-factor authentication is a security method that adds a second layer of identity verification. Instead of relying on just a password, 2FA requires you to confirm your identity using two distinct factors. This reduces the risk of unauthorized access, even if a password is compromised.

How it works???

The 2FA process it adds a real time verification step to standard login workflow.

1. We enter our username and password on any site or apps.
2. The server recognizes the password and request the second factor.
3. We complete the 2^{nd} factor by entering a otp , or a hardware key or approve a push notification.
4. If the server approves or validates this second factor , you can access the app or the site.

Why it exists??

Basically the reason 2FA exists is to stop unauthorized access, particularly in cases where passwords have been leaked or stolen via data breaches. It exists to

- Prevent account takeover
- To protect against phising
- And to reduce automated attacks

Passwords alone are very weak and they are not very sufficient nowadays to secure digital accounts particularly in this big year of 2026 where it is really easy to steal, guess or reuse the same passwords.

Phishing and social engineering, credential stuffing, data breaches, ai-powered attacks, malwares and keyloggers are few examples of the cyber attacks we can face.

Types of 2FAs:

- S**MS OTP Authentication:** it authenticates the user through a one time code(OTP) send by sms or emails.

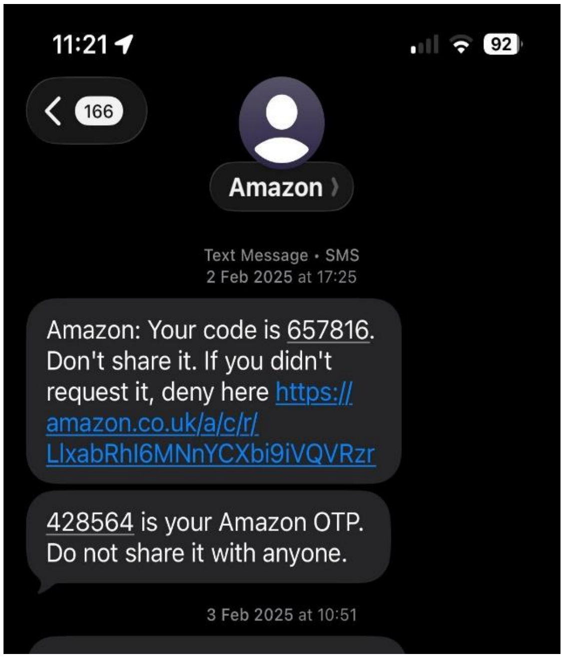
- 

Figure 1example of otp

- **authenticator Apps**: the authenticator apps enhance security over sms, with many providing encrypted backups and multi-device syncing.

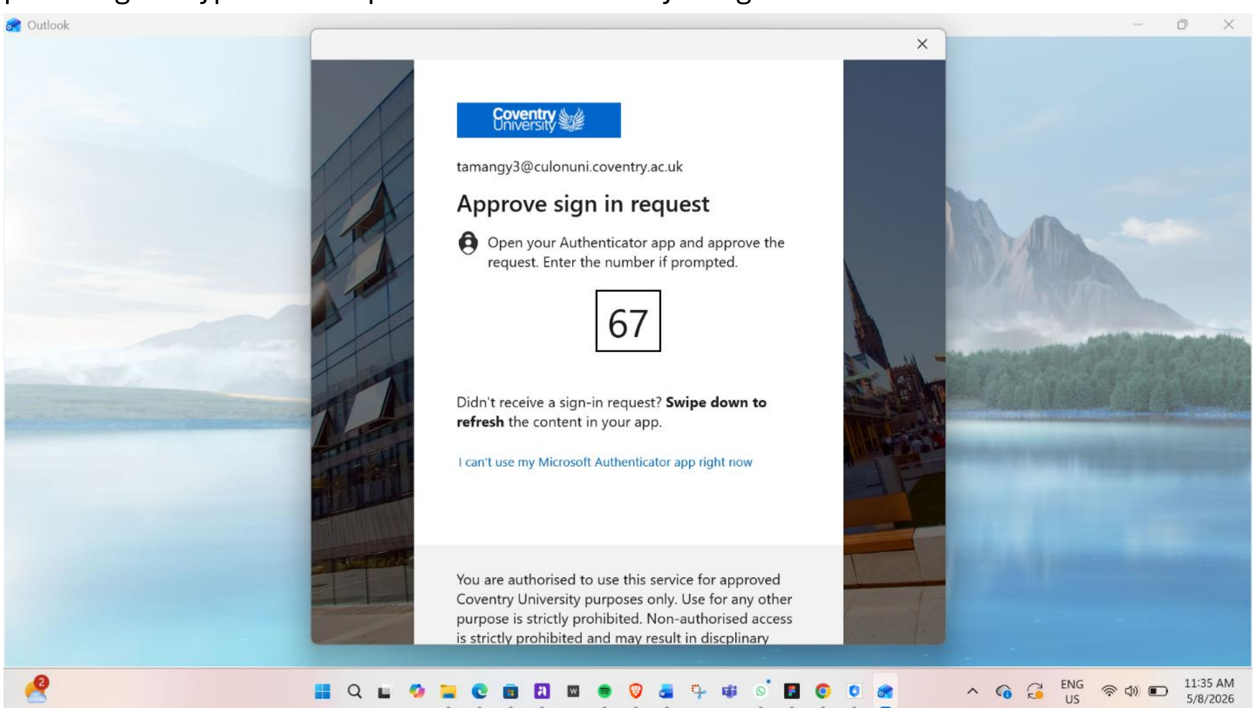


Figure 2 example authenticator app

- **Email verification**: 2FA email verification examples include concise, time-sensitive codes or magic links sent to a user's registered email after they enter their password

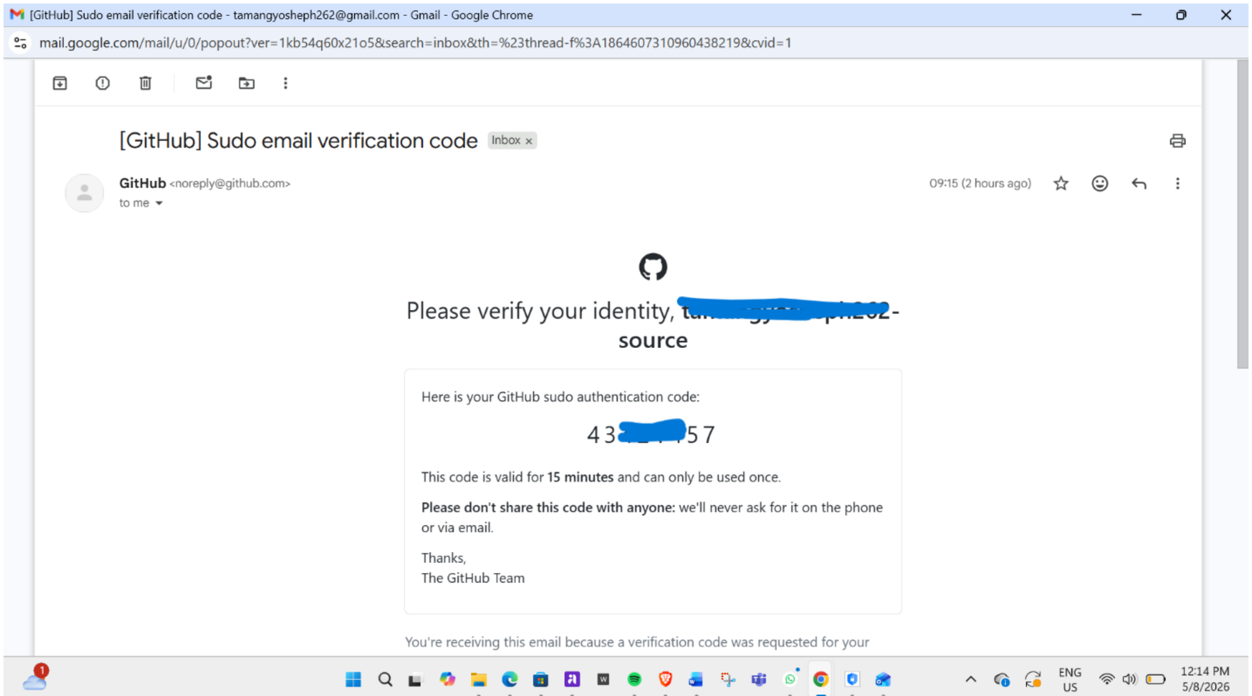


Figure 3 example of email verification

- **Push notification:** it provide a secure, one tap method to authorize logins by sending an instant prompt to a users mobile device.

Hi Yosheph,

Someone just logged into your Facebook account near Leeds, United Kingdom on Facebook for iPhone. If this wasn't you, we're here to help you take some simple steps to secure your account.

Was this you?

Leeds, United Kingdom

Facebook for iPhone

Thursday, April 9, 2026 at 3:54 PM

This wasn't me

If this was you, you can ignore this email.

Wondering if this email is really from us? Visit the Help Center to confirm:
www.facebook.com/help/check-email

To improve your account security, we've enabled login alerts. We'll continue to notify you whenever your username and password are used to login from a new browser or device.

Thanks,
Facebook Security

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- **Biometrics:** Biometric 2FA (Two-Factor Authentication) uses unique physical or behavioral characteristics, "something you are" as the second layer of security after a password. Common examples include fingerprint scans, facial recognition (FaceID), iris scans, and voice recognition. These methods provide high security and convenience for accessing devices and apps

Comaprision between 2FA methods

2FA method	How it works	Weaknesses	Real-life use
SMS OTP	Code send by text message	SIM swap scams, weak signals	Instagram, banking apps
Authenticator apps	App generates login code	Phone loss can lock users out	Gmail, discord, GitHub

Email verification	Code link send to email	Unsafe if email is hacked	Password resets, logins
Push notifications	User approves login	MFA fatigue spam attacks	Microsoft, banking apps
biometrics	Fingerprint or face ID login	Sensor errors, privacy concerns	Smartphones, banking apps

Common cyber threats in 2FA:

- **Adversary-in-the-Middle (AiTM) Phising**: the attackers create realistic fake websites or phishing pages that acts as a proxy between the user and the legitimate service. What usually happens is that the user will enter their username, password and 2FA code into the fake website allowing the attackers script to catch the credentials gets access to the services.
- MFA Fatigue